Consistent sets without identity have denumerable models

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openlogic:first-order-logic/completeness:local-snapshot-20261013;
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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel noidentity-ls of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the completeness theorem for first-order logic without identity and the term model it constructs
- the Henkin construction delivering a structure whose domain is the closed terms of the extended language
- denumerability of the term set of a countable language

Theorem 0.1. *If Γ is a consistent set of !!sentences in the language of first-order logic without identity, then it has !!adenumerable model, i.e., it is satisfiable in !!astructure whose domain is infinite and !!enumerable.*

Proof. If Γ is consistent and contains no sentences in which identity appears, then the !!structure $\mathfrak{A}M$ delivered by the proof of the completeness theorem has a domain $|\mathfrak{A}|M$ identical to the set of terms of the language $\mathcal{L}L'$. So $\mathfrak{A}M$ is !!denumerable, since $\mathrm{Trm}[L']$ is. □

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